EYETHEEKH S P

Kochi, Kerala – eyetheekhsp@gmail.com – +91 9995581029 – linkedin.com/in/eyetheekh – gitlab.com/eyetheekh

PROFILE

Python Developer with nearly **2 years** of experience in **Backend Engineering**, **Microservices**, and **AI/ML** applications. Skilled in building **asynchronous APIs**, integrating **GenAI/ML pipelines**, and deploying scalable systems. Proficient in designing and developing **containerized applications**, **microservices**, and **distributed workflows**. Strong emphasis on **clean code**, security compliance and **system design**.

EXPERIENCE

Jr Python Developer **June 2024 – Present**
PixDynamics Pvt Ltd Kochi, Kerala

- Designed, developed, and maintained **production-ready backend services** with a focus on clean API design, validation, security, and performance.
- Built **asynchronous REST APIs**, **gRPC** services, and **background workers** using **Celery, Dramatiq, Redis, and custom workers** to handle large-scale workloads.
- Implemented **event-driven** and **message-based architectures** using **Kafka, RabbitMQ,** and **Redis**, enabling scalable task orchestration, decoupled services, and reliable inter-service communication.
- Integrated numerous **machine learning and GenAI pipelines** into backend services, improving concurrency, throughput, and API response times by applying **async** best practices and memory-aware processing.
- Containerized applications using **Docker** multi-stage builds, creating custom hardened base images to **reduce image size, attack surface, and security vulnerabilities** for production deployments.
- Worked extensively on **Linux-based production environments**, troubleshooting issues related to networking, dependencies, resource utilization, and container runtime behavior.
- Actively contributed across the full development lifecycle, including **system design, development, testing, optimization, and deployment**.

Python Developer Intern **September 2023 - April 2024**
Luminar Technolab Kochi, Kerala

- Developed full-stack applications using **Django, FastAPI, and React**, and deployed containerized solutions to **AWS** and **GCP**, gaining hands-on expertise in cloud infrastructure and **CI/CD** pipelines.

PROJECTS

Aadhaar Masking Bulk Processor

- Developed and deployed a scalable **asynchronous microservice system** on **AWS** to mask sensitive data in **4+ crore documents** within a 2-month deadline.
- Implemented a **message queue-based architecture** using **Redis**, **Postgres+SQLAlchemy** for job orchestration, distributed worker pool to handle concurrent processing, resulting in efficient task distribution, leveraging **gRPC** to isolate ML inference and prevent resource-heavy model processing from impacting core service stability.
- Developed an automated reporting utility to track processing metrics and export error logs into Excel for operational visibility.

OVD (Officially Valid Document) Extractor

- Maintained and improved FastAPI-based machine learning and computer vision pipelines for document extraction (Passport, PAN, Aadhaar, Voter ID).
- Migrated the monolithic codebase into **gRPC**-based microservices to isolate ML models and OCR inference, improving scalability and concurrency.
- Improved application security by performing vulnerability scans using Trivy, creating reusable custom **Docker** base images to reduce image size and attack surface, and securing file upload endpoints with YARA-based scanning to detect malicious content.
- Implemented **automated testing** with **Pytest** and performed load and stress testing using **Locust** and **Grafana k6** to fix memory leaks, optimize resource usage, improve concurrency, and enhance API response times.

AI-Driven Document to Structured JSON Extractor

- Built and deployed a fully asynchronous, horizontally scalable pipeline to convert PDF documents from **MinIO S3** into structured **JSON** using **GenAI** for high-accuracy data extraction.
- Implemented an asynchronous microservice architecture using **Celery**, **Redis**, and **MinIO S3**, with retries, webhooks, and scheduler for dead-letter queue replay to improve reliability.

Multi-Modal AI Chatbot

- Implemented secure **JWT** authentication with database-backed user management and encrypted credential storage.
- Implemented **Kafka**-based background tracking system for token usage and cost monitoring.

Account Opening Form Validator

- Developed a backend service to validate account opening forms using rule-based logic and integrated **ML-based checks**, reducing manual review effort.
- Implemented **job orchestration**, **state tracking**, and caching with **Redis**, and integrated **MinIO S3** for result storage and sharing.
- Strengthened application security by adding **YARA**-based malware detection for user-uploaded files.

Multi-LLM Gateway SDK

- Developed a Python SDK that provides a unified interface for interacting with multiple GenAI providers, including **OpenAI**, **Google Generative AI**, **HuggingFace**, **Ollama**, and **vLLM**.
- Simplified provider switching and reduced vendor lock-in by abstracting provider-specific logic behind a common API supporting multimodal inputs and synchronous, asynchronous, and streaming responses.
- Integrated a **Redis** cache layer to minimize database overhead and reduce latency.

API Framework

- Built and maintained a standardized, reusable backend framework using **FastAPI**, with built-in authentication, request validation, multiple database support, testing, and API metrics to speed up development and deployment.
- Added cloud storage support for file access and downloads, including **Google Drive (OAuth 2.0)** and **Microsoft OneDrive (MSAL)**.

SKILLS

- **Languages:** Python, JavaScript, SQL
- **Backend & Frameworks:** FastAPI, Django, Django REST Framework, Flask, aiohttp, gRPC, GraphQL
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, Redis, SQLite, MinIO
- **Messaging & Streaming:** Kafka, RabbitMQ, Redis
- **DevOps & Cloud:** Docker, Docker Compose, AWS, Linux, Nginx, CI/CD, Trivy
- **Testing & Monitoring:** Pytest, Locust, Grafana k6, Postman
- **Version Control & Collaboration:** Git, GitLab
- **Architecture & Concepts:** Microservices, Event-Driven Architecture, AsyncIO, System Design
- **Professional Skills:** Debugging, Problem Solving

CERTIFICATIONS

- **IBM Back-End Development Professional Certificate** - IBM
- **IBM Applied Software Engineering Fundamentals Specialization Certificate** - IBM
- **AWS Cloud Technical Essentials** - AWS
- **META Programming in Python** - META
- **Introduction to Linux LFS101** - The Linux Foundation

EDUCATION

B.Com Computer Applications — University of Kerala, 2020 – 2023